



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

AIOT

14-08-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE ADVANCED)

ANSWER KEY

PAPER-1

PART-1 : PHYSICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	A	B	A	A	A
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C,D	A,C	A,B,C	B,D	B,C	A,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	55.00	2.00	1.10	1.60	3.74	26.00

PART-2 : CHEMISTRY

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	D	B	B	C	D	B
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,C	A,B,D	A,B,D	B,C,D	B,C,D	A,B,C
SECTION-II	Q.	1	2	3	4	5	6
	A.	5.00	7.00	4.00	2.00	1.00	8.00

PART-3 : MATHEMATICS

SECTION-I (i)	Q.	1	2	3	4	5	6
	A.	C	B	C	B	C	C
SECTION-I (ii)	Q.	7	8	9	10	11	12
	A.	A,B,D	A,C,D	B	C,D	A,C,D	A,B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	7.34 or 7.35	0.25	2.65 or 2.66	9.00	55.78	5.00

HINT – SHEET

PART-1 : PHYSICS

SECTION-I (i)

1. Ans (C)

$$I = I_0 \sin^2 \theta$$

Differentiating I with respect to θ

$$\frac{dI}{d\theta} = 2I_0 \sin \theta \cos \theta$$

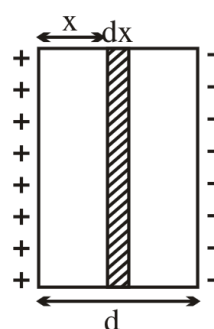
$$\frac{dI}{I} = \frac{2I_0 \sin \theta \cos \theta d\theta}{I_0 \sin^2 \theta} = 2 \cot \theta d\theta$$

Percentage error in angle

$$= \frac{d\theta}{\theta} \times 100 = \left(\frac{dI}{2I \cot \theta} \right) \frac{I}{\theta} \times 100$$

$$= \frac{0.002}{2 \times 5 \cot 30^\circ} \times \frac{6 \times 100}{\pi} = \frac{4}{\pi} \sqrt{3} \times 10^{-2} \%$$

2. Ans (A)



$$K(x) = \frac{K_2 - K_1}{d} x + K_1$$

$$\frac{dK}{dx} = \left(\frac{K_2 - K_1}{d} \right)$$

$$\text{Now, } dC = \frac{K \epsilon_0 A}{dx}$$

$$\Rightarrow \frac{1}{C_{eq}} = \int \frac{dx}{K \epsilon_0 A}$$

$$\Rightarrow \frac{1}{C_{eq}} = \frac{d}{K_2 - K_1} \int_{K_1}^{K_2} \frac{dK}{K \epsilon_0 A}$$

$$\Rightarrow \frac{1}{C_{eq}} = \frac{d}{(K_2 - K_1) \epsilon_0 A} \ln \left(\frac{K_2}{K_1} \right)$$

$$\Rightarrow C_{eq} = \frac{(K_2 - K_1) \epsilon_0 A}{d \ln \left(\frac{K_2}{K_1} \right)}$$

3. **Ans (B)**

Since both particles have same magnitude of momentum both have same radius.

$$r = \frac{p}{qB}$$

But since masses are different time periods will be different.

$$T = \frac{2\pi m}{qB}$$

Hence both meet after sweeping different angles.

4. **Ans (A)**



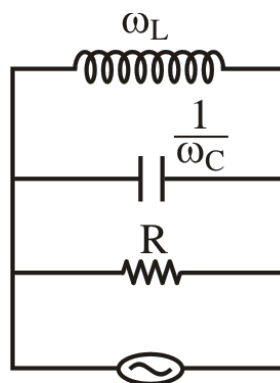
$$\bullet \longrightarrow E = \frac{\sigma}{2\epsilon_0}$$

$$dU = dV = -Edx$$

$$\Rightarrow W_e = \Delta U_e$$

$$= - \int_{x_1}^{x_2} E dx = \frac{-\sigma}{2\epsilon_0} \int_{x_1}^{x_2} dx = \frac{\sigma}{2\epsilon_0} (x_1 - x_2)$$

5. **Ans (A)**



$$v = \epsilon_0 \sin \omega t$$

Peak current through source

$$I_0 = \frac{\epsilon_0}{Z}$$

Peak current through resistor

$$I_R = \frac{\epsilon_0}{R} = \frac{1}{2} \frac{\epsilon_0}{Z}$$

$$\Rightarrow R = 2Z$$

$$\frac{1}{Z} = \frac{1}{R^2} + \left(\frac{1}{X_C} - \frac{1}{X_L} \right)^2$$

$$\Rightarrow \frac{4}{R^2} = \frac{1}{R^2} + \left(\omega C - \frac{1}{\omega L} \right)^2$$

$$\Rightarrow \frac{3}{R^2} = \left(\omega C - \frac{1}{\omega L} \right)^2$$

$$\Rightarrow R = \frac{\sqrt{3}}{\left| \omega C - \frac{1}{\omega L} \right|}$$

6. **Ans (A)**

$$\left(\frac{IA \cos \theta}{hf} \right) \left(\frac{hf}{c} 2 \cos \theta \right) = kx$$

$$\frac{2IA \cos^2 \theta}{ck} = x = 0.04m$$

PART-1 : PHYSICS

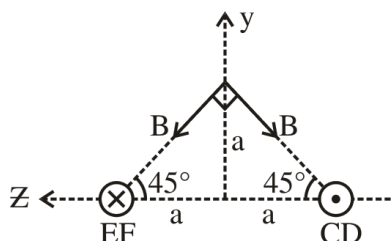
SECTION-I (ii)

7. Ans (A,B,C,D)

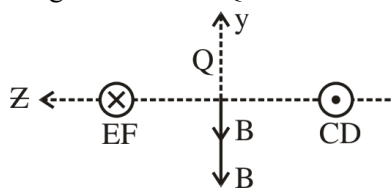
Here in all case we just have to indicate direction.

Magnetic field at P due to CD & EF.

Resultant will be along $-\hat{j}$.

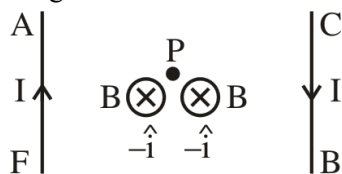


Magnetic field at Q due to EF and CD.

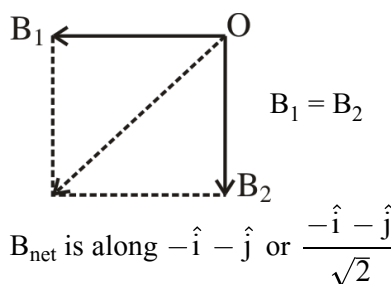


Resultant will be along $-\hat{j}$.

Magnetic field at P due to AF and CB



At O magnetic field due to AF and CB is along $-\hat{i}$ and due to EF and CD is along $-\hat{j}$.



9. Ans (A,B,C)

$$\lambda_B = 2\lambda_A$$

$$\Rightarrow \frac{\lambda_B}{\lambda_A} = 2 = \sqrt{\frac{T_A}{T_B}}$$

$$\Rightarrow T_A = 4T_B$$

$$\Rightarrow T_A = 4(T_A - 1.5) = 4T_A - 6$$

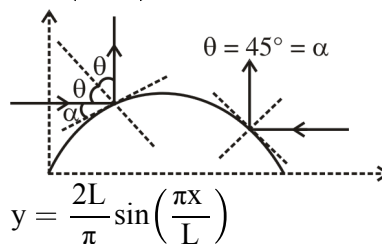
$$\Rightarrow 3T_A = 6 \Rightarrow T_A = 2 \text{ eV}$$

$$\text{and } T_B = 0.5 \text{ eV}$$

$$\phi_A = 4.25 - T_A = 2.25 \text{ eV}$$

$$\phi_B = 4.70 - T_B = 4.20 \text{ eV}$$

10. Ans (B,D)



Slope of the curve = ± 1 at the point of incidence

$$\frac{dy}{dx} = \frac{2L}{\pi} \cos\left(\frac{\pi x}{L}\right) \times \frac{\pi}{L} = 2 \cos\left(\frac{\pi x}{L}\right)$$

Case-I

$$\Rightarrow 2 \cos\left(\frac{\pi x}{L}\right) = 1$$

$$\Rightarrow \cos\left(\frac{\pi x}{L}\right) = \frac{1}{2}$$

$$\Rightarrow \frac{\pi x}{L} = \frac{\pi}{3} \Rightarrow x = \frac{L}{3}$$

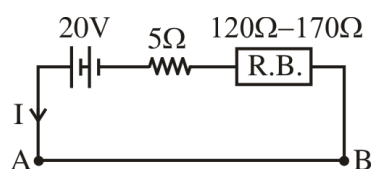
$$\text{and } y = \frac{2L}{\pi} \sin\left(\frac{\pi}{3}\right) = \frac{L\sqrt{3}}{\pi}$$

Case-II

$$2 \cos\left(\frac{\pi x}{L}\right) = -1$$

$$\Rightarrow \frac{\pi x}{L} = \frac{2\pi}{3} \Rightarrow x = \frac{2L}{3}, y = \frac{L\sqrt{3}}{\pi}$$

11. Ans (B,C)



$$I_{\min} = \frac{20}{75 + 5 + 170} = \frac{20}{250} = \frac{2}{25} \text{ A}$$

$$V_{AB} \text{ in this case} = \frac{2}{25} \times 75 = 6 \text{ V}$$

$$I_{\max} = \frac{20}{75 + 5 + 120} = \frac{20}{200} = \frac{1}{10} \text{ A}$$

$$V_{AB} \text{ in this case} = \frac{1}{10} \times 75 = 7.5 \text{ V}$$

Hence voltages between 6V and 7.5V can be measured.

12. Ans (A,C,D)

$$T_A = \frac{T_0}{2^{\gamma-1}}; \quad T_C = 2T_0; \quad T_D = \frac{T_0}{2^{\gamma-2}};$$

$$P_A = \frac{P_0}{2^{\gamma}}; \quad P_D = \frac{P_0}{2^{\gamma-1}}$$

$$\Delta Q = C_V(T_C - T_B) + C_V(T_A - T_D)$$

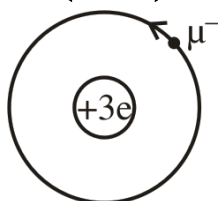
$$= C_V T_0 \left(1 - \frac{1}{2^{\gamma-1}}\right)$$

$$\Delta W = \Delta Q$$

PART-1 : PHYSICS

SECTION-II

1. Ans (55.00)



$$E_n = \frac{-me^4 Z^2}{8\epsilon_0^2 h^2 n^2}$$

$$E_n = \frac{-208 m_e e^4 Z^2}{8\epsilon_0^2 h^2 n^2}$$

$$E_n = -208 \times 13.6 \frac{Z^2}{n^2}$$

$$\Delta E = 208 \times 13.6 Z^2 \left(\frac{1}{1^2} - \frac{1}{3^2} \right)$$

$$\frac{hc}{\lambda} = 208 \times 13.6 \times 9 \times \frac{8}{9} \quad \{Z=3\}$$

$$\Rightarrow \lambda = \frac{1.24 \times 10^{-6}}{208 \times 13.6 \times 8} = 5.49 \times 10^{-11} \approx 55 \text{ pm}$$

2. Ans (2.00)

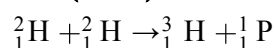
$$\varepsilon = L_1 \frac{di}{dt} - M_{12} \frac{di}{dt} + L_2 \frac{di}{dt} - M_{21} \frac{di}{dt}$$

$$= (L_1 + L_2 - 2M) \frac{di}{dt}$$

$$= (1 + 1 - 1) (2)$$

$$= 2 \text{ volt}$$

3. Ans (1.10)



Using momentum conservation

$$\vec{P}_1 + \vec{P}_2 = 0$$

$$\vec{P}_1 = -\vec{P}_2$$

$$P_1^2 = P_2^2$$

$$2m_1 k_1 = 2m_2 k_2$$

$$3k_1 = k_2 \quad \dots (i)$$

Also,

$$k_1 + k_2 = 8.4 - 4$$

$$k_1 + k_2 = 4.4 \quad \dots (ii)$$

solving (i) and (ii) we get

$$4k_1 = 4.4$$

$$k_1 = 1.1 \text{ MeV and } k_2 = 3.3 \text{ MeV}$$

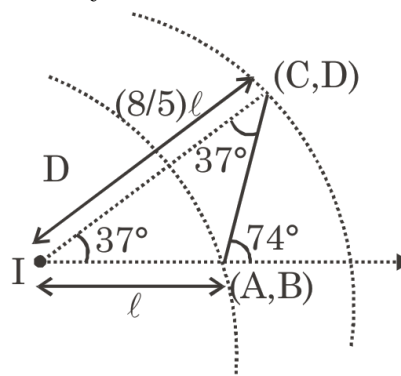
kinetic energy of the tritium, $k_1 = 1.1 \text{ MeV}$

kinetic energy of the proton, $k_2 = 3.3 \text{ MeV}$

4. Ans (1.60)

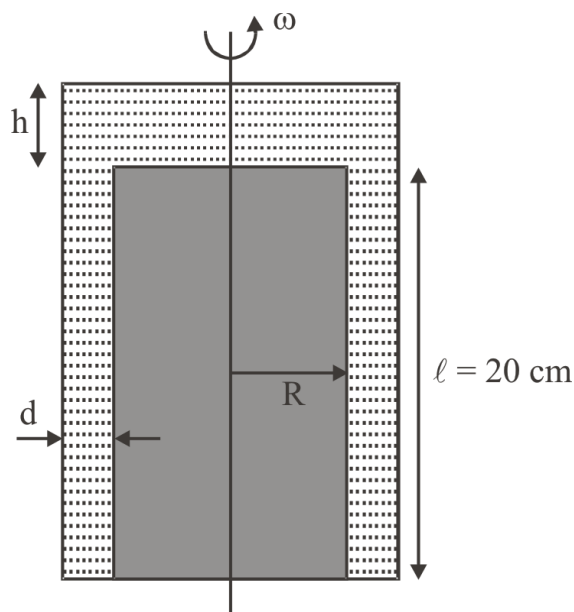
Flux passing through the loop ABCD

$$\phi = \int_{\ell}^{(8/5)\ell} \frac{\mu_0 I}{2\pi r} \ell dr = \frac{\mu_0 I \ell}{2\pi} \ln \left(\frac{8}{5} \right)$$



5. Ans (3.74)

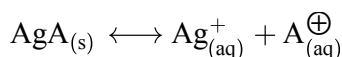
The net torque of viscous force acting on the cylinder.



$$\begin{aligned} \tau &= \int_0^R \eta 2\pi r dr \frac{\omega r}{h} + \eta 2\pi \ell \frac{\omega R}{d} R \\ &= \frac{\pi \eta R^4 \omega}{2h} + \frac{2\pi \eta R^3 \ell \omega}{d} = \pi \eta R^3 \left(\frac{R}{2h} + \frac{2\ell}{d} \right) \omega \\ &= \frac{22}{7} \times 7 \times 10^{-6} \left(\frac{10}{0.2} + \frac{40}{0.05} \right) 50 \end{aligned}$$

$$t = 22 \times 10^{-6} \times 850 \times 20 = 0.374 \text{ N-m}$$

Saturated solution

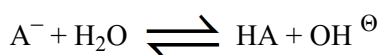


$$[\text{Ag}^{+}][\text{A}^{-}] = K_{sp}$$

$$S = [\text{Ag}^{+}] = [\text{A}^{-}] \therefore \text{No external source of } [\text{A}^{\ominus}]$$

S = solubility of silver salt

Hydrolysis of $[\text{A}^{-}]$



S

$$S(1-\alpha) \quad S\alpha \quad S\alpha$$

$$K_h = \frac{K_w}{K_a} = \frac{S\alpha^2}{1-\alpha} = 10^{-4}$$

assuming $\alpha \ll 1$ since $\text{pH} = 9$; $\text{pOH} = 5$

$$[\text{OH}^{-}] = \sqrt{K_h \cdot S} = 10^{-5} \quad [\text{OH}^{-}] = 10^{-5}$$

$$S = 10^{-6}$$

$$(\text{OH}^{-}) = S\alpha = 10^{-5}$$

$\alpha = 0.1$ not negligible

assumption $\alpha \ll 1$ is incorrect

$$K_h = \frac{[\text{HA}][\text{OH}^{-}]}{[\text{A}^{-}]} = 10^{-4} = \frac{10^{-10}}{S - 10^{-5}}$$

$$S - 10^{-5} = 10^{-6} \quad 1.1 \times 10^{-5} \times \alpha = 10^{-5}$$

$$S = 1.1 \times 10^{-5} \quad \alpha = \frac{1}{1.1} = \frac{10}{11}$$

$$s(1-\alpha)S = K_{sp}$$

$$1.1 \times 10^{-5} \times 1.1 \times 10^{-5} \left(1 - \frac{10}{11}\right) = K_{sp}$$

$$1.1 \times 10^{-11} = K_{sp} \text{ option (c) is correct}$$

PART-2 : CHEMISTRY

SECTION-I (ii)

8. Ans (A,B,D)

(A) $\text{HF} < \text{HCl} < \text{HBr} < \text{HI}$ (Bond length)

top to bottom 'n' increases, no. of shell increases as size increases. bond length $\uparrow$

(B) More Z effective causes more polarisation

(C) $\text{Li} < \text{Be} < \text{B} < \text{C}$ (IE_1).

correct order $\text{C} > \text{Be} > \text{B} > \text{Li}$

$$\text{Li} : 1s^2 2s^1$$

$$\text{Be} : 2s^2 2s^2$$

$$\text{B} : 1s^2 2s^2 2p^1$$

$$\text{C} : 1s^2 2s^2 2p^2$$

(D) $\text{Li}^{+}(\text{g}) < \text{Na}^{+}(\text{g}) < \text{K}^{+}(\text{g}) < \text{Cs}^{+}(\text{g})$ (Ionic Radius)

In a group top to bottom size $\uparrow$

Ans. A,B,D.

10. Ans (B,C,D)

(A) $Z > 1$ if repulsive force dominates gas is less compressible.

(B) $Z = 1$ [at very low P and high T]

(C) $Z = 1$ at Boyle's temperature for a range of pressure.

(D) Fact.

11. Ans (B,C,D)

(A) $\text{H} - \text{H}(\text{g}) \rightarrow 2\text{H}(\text{g})$

B.E. of $\text{H} - \text{H} = 2 \times \Delta H_f$

(B) I.E. of Na : $\text{Na} \rightarrow \text{Na}^{+} + \text{e}^{-}$

ΔH_{eg} of Na^{+} : $\text{Na}^{+} + \text{e}^{-} \rightarrow \text{Na}$

$\therefore \Delta H_{eg} = -E.A.$

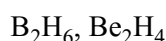
(C) Already given

(D) $|\Delta H_{comb.}| \propto \frac{1}{\text{stability}}$

PART-2 : CHEMISTRY

SECTION-II

4. Ans (2.00)



5. **Ans (1.00)**

$$\lambda = \frac{h}{\sqrt{3mkT}}, \lambda \propto m^{-1/2} T^{-1/2}$$

6. **Ans (8.00)**

$$\Delta S_{\text{water}} = \frac{3000}{300} = 10 \text{ J/K}$$

$$\Delta S_{\text{fish}} = \frac{-2}{3} \times 10^{-3} \times 3000 = -2 \text{ J/K}$$

$$\Delta S_{\text{total}} = 8 \text{ J/K}$$

PART-3 : MATHEMATICS

SECTION-I (i)

1. **Ans (C)**

$$f(x) = 4x^2 + \frac{\alpha}{x} \Rightarrow f'(x) = 8x - \frac{\alpha}{x^2} = \frac{8x^3 - \alpha}{x^2}$$

$$\therefore f(x) \text{ attains its minimum at } x = \left(\frac{\alpha}{8}\right)^{1/3}$$

$$\Rightarrow f\left(\left(\frac{\alpha}{8}\right)^{1/3}\right) = 4\left(\frac{\alpha}{8}\right)^{2/3} + \frac{\alpha}{\left(\frac{\alpha}{8}\right)^{1/3}} = (3)^{1/3}$$

$$\Rightarrow \alpha = \frac{1}{3}$$

2. **Ans (B)**

$$\frac{\sin^2 \theta \cdot \sin \theta}{\cos \theta} + \frac{\cos^3 \theta}{\sin \theta} = (1 + \sin 2\theta) + \frac{(\sin^2 \theta + \cos^2 \theta)}{\sin \theta \cdot \cos \theta}$$

$$\Rightarrow \frac{\sin^4 \theta + \cos^4 \theta}{\sin \theta \cdot \cos \theta} = (1 + \sin 2\theta) + \frac{1}{\sin \theta \cdot \cos \theta}$$

$$\Rightarrow \frac{1 - 2\sin^2 \theta \cos^2 \theta}{\sin \theta \cdot \cos \theta} = (1 + \sin 2\theta) + \frac{2}{(\sin 2\theta)}$$

$$\Rightarrow \frac{2\left(1 - \frac{\sin^2 2\theta}{2}\right)}{\sin 2\theta} = (1 + \sin 2\theta) + \frac{2}{\sin 2\theta}$$

$$\Rightarrow \frac{2\left(1 - \frac{t^2}{2}\right)}{t} = (1 + t) + \frac{2}{t}$$

$$\Rightarrow \frac{(2 - t^2)}{t} = \frac{t^2 + t + 2}{t}$$

$$\Rightarrow 2t^2 + t = 0 \Rightarrow t(2t + 1) = 0$$

$$\Rightarrow t = 0 \quad \& \quad t = -\frac{1}{2}$$

$$\sin 2\theta = 0$$

$$\sin \theta = 0; \cos \theta = 0 \quad \sin 2\theta = -\frac{1}{2}$$

3. **Ans (C)**

$$P(T_1) = \frac{20}{100} \quad P(T_2) = \frac{60}{100} \quad P(T_3) = \frac{20}{100}$$

$$\text{Let } P\left(\frac{D}{T_2}\right) = x$$

$$P\left(\frac{D}{T_1}\right) = 10x = P\left(\frac{D}{T_3}\right)$$

$$P(D) = \frac{10}{100} \quad (\text{given})$$

$$P(T_1)P\left(\frac{D}{T_1}\right) + P(T_2)P\left(\frac{D}{T_2}\right) + P(T_3)P\left(\frac{D}{T_3}\right) = \frac{1}{10}$$

$$\frac{20}{100} \times 10x + \frac{60}{100} \times x + \frac{20}{100} \times 10x = \frac{1}{10}$$

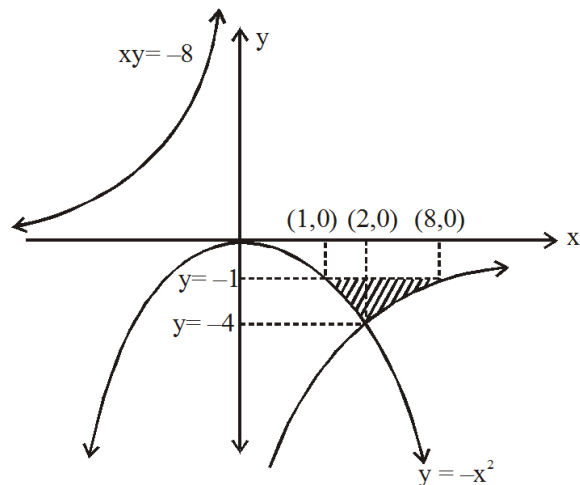
$$\Rightarrow x = \frac{1}{46}$$

$$P\left(\frac{D}{T_2}\right) = \frac{1}{46} \Rightarrow P\left(\frac{\bar{D}}{T_2}\right) = \frac{45}{46}$$

$$P\left(\frac{D}{T_3}\right) = P\left(\frac{D}{T_1}\right) = \frac{10}{46} \Rightarrow P\left(\frac{\bar{D}}{T_1}\right) = \frac{36}{46} = P\left(\frac{\bar{D}}{T_3}\right)$$

$$P\left(\frac{T_2}{\bar{D}}\right) = \frac{\frac{20}{100} \times \frac{36}{46} + \frac{60}{100} \times \frac{45}{46} + \frac{20}{100} \times \frac{36}{46}}{\frac{1}{10}} = \frac{15}{23}$$

4. **Ans (B)**



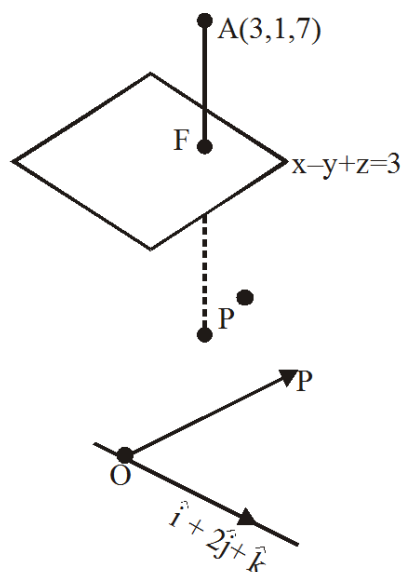
Required Region =

$$A = \int_1^2 (-1 - (-x^2)) dx + \int_2^8 \left(-1 - 1\left(-\frac{8}{x}\right)\right) dx$$

$$A = 16\ln 2 - \frac{4}{3}$$

5. Ans (C)

Line AP : $\frac{x-2}{1} = \frac{y-1}{-1} = \frac{z-5}{1} = \lambda$



$\Rightarrow F(2+\lambda, 1-\lambda, \lambda+5)$ lies in the plane

$\therefore 2+\lambda - (1-\lambda) + \lambda + 5 = 3$

$3\lambda = -3 \Rightarrow \lambda = -1$

$\Rightarrow F(1,2,4)$

$\Rightarrow P(0,3,3)$

so required plane is $\begin{vmatrix} x-1 & y-2 & z-3 \\ 1 & 2 & 1 \\ -1 & 1 & 0 \end{vmatrix} = 0$

$\therefore x + y - 3z + 6 = 0$

6. Ans (C)

We have,

$$\begin{aligned} &= 2 \sum_{k=1}^{13} \frac{\sin\left(\left(\frac{k\pi}{6} + \frac{\pi}{4}\right) - \left((k-1)\frac{\pi}{6} + \frac{\pi}{4}\right)\right)}{\cos\left(\frac{\pi}{4} + (k-1)\frac{\pi}{6}\right) \cdot \cos\left(\frac{\pi}{4} + \frac{k\pi}{6}\right)} \\ &= 2 \sum_{k=1}^{13} \tan\left(\frac{k\pi}{6} + \frac{\pi}{4}\right) - \tan\left((k-1)\frac{\pi}{6} + \frac{\pi}{4}\right) \\ &= 2 \left[\tan\left(\frac{13\pi}{6} + \frac{\pi}{4}\right) - \tan\left(\frac{\pi}{4}\right) \right] \\ &= 2 [2 + \sqrt{3} - 1] = 2(1 + \sqrt{3}) \end{aligned}$$

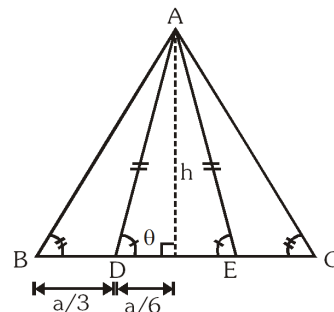
PART-3 : MATHEMATICS

SECTION-I (ii)

7. Ans (A,B,D)

$\therefore AD = AE$ & $BD = EC$

$\therefore AB = AC$



$\Rightarrow \triangle ABC$ is isosceles

$\Rightarrow \angle B = \angle C$

$\tan B = \frac{h}{a/2} = \frac{2h}{a}$

$\tan \theta = \frac{h}{a/6} = \frac{6h}{a}$

$\Rightarrow \tan \theta = 3 \tan B$

$\Rightarrow 3 \tan B = \tan(\pi - \angle ADB)$

$\Rightarrow 3 \tan B + \tan(\angle ADB) = 0$

Similarly $3 \tan C + \tan(\angle AEC) = 0$

$A + B + C = \pi$

$2B = \pi - A$

$\tan 2B = -\tan A \Rightarrow \tan A + \tan 2B = 0$

8. Ans (A,C,D)

(A) $f'(x) = 3x^2 + 2$

$g'(f(x)) \cdot f'(x) = 2 \Rightarrow g'(f(x)) = \frac{2}{f'(x)}$

$\Rightarrow g'(2) = \frac{2}{f'(0)} = \frac{2}{2} = 1$

(B) $h(g(g(x))) = 5x + 7$

$h'(g(g(x))) = \frac{5}{g'(g(x)) \cdot g'(x)}$

Now, $g(g(x)) = 5$

$g(x) = f(1) = 5$

$\therefore x = f(1) = 5$

so $h'(5) = \frac{5}{g'(5) \cdot g'(5)} = \frac{5}{f'(1) \cdot f'(1)} = \frac{5}{2 \times 2} = \frac{125}{4}$

(C) $g(g(x)) = 5$

$\therefore g(x) = g^{-1}(5) \Rightarrow g(x) = f(1) \Rightarrow g(x) = 5$

$\Rightarrow x = g^{-1}(5) \Rightarrow x = f(1) \Rightarrow x = 5$

so $h(5) = 5 \times 5 + 7 = 32$

(D) $g(x) = 3 \Rightarrow x = g^{-1}(3) \Rightarrow x = f(0) \Rightarrow x = 2$

so $h(g(3)) = 5 \times 2 + 7 = 17$

9. Ans (B)

$$PQ = kI$$

$$|P| \cdot |Q| = k^3$$

$$\Rightarrow |P| = 2k \neq 0 \Rightarrow P \text{ is an invertible matrix}$$

$$\therefore PQ = kI$$

$$\therefore Q = kP^{-1}I$$

$$\therefore Q = \frac{\text{adj. } P}{2}$$

$$\therefore q_{23} = -\frac{k}{8}$$

$$-\frac{2\alpha + 6}{2} = -\frac{k}{8} \Rightarrow k = 8\alpha + 24 \quad \dots (1)$$

$$|P| = 2k \Rightarrow k = -3 - 4\alpha \quad \dots (2)$$

$$\text{from (1) and (2) } \alpha = -\frac{9}{4} \text{ and } k = 6$$

$$\therefore 4\alpha - k + 8 = 0$$

$$\begin{aligned} \& \det(P(\text{adj. } Q)) = |P| |\text{adj. } Q| = 2k \cdot \left(\frac{k^2}{2}\right)^2 \\ &= \frac{k^5}{2} = \frac{(6)^5}{2} = 2^4 \times 3^5 \end{aligned}$$

10. Ans (C,D)

$$\lambda x + 2y = a$$

$$-\mu x + 2y = -3$$

$$\text{if } \lambda = -\mu \text{ and } a = -3$$

$$\text{then lines are coincident and } \lambda = -\mu \text{ and } a \neq -3$$

$$\text{then lines are parallel}$$

11. Ans (A,C,D)

$$f(x) = \begin{cases} 1 & x \in (2n\pi, 2n\pi + \frac{\pi}{2}) & n \in \mathbb{I}^+ \cup \{0\} \\ \frac{1+5\log_e x}{3} & x = (4n+1)\frac{\pi}{2} & n \in \mathbb{I}^+ \cup \{0\} \\ \frac{5\log_e x}{2} & x \in (2n\pi + \frac{\pi}{2}, 2n\pi + \frac{3\pi}{2}) & n \in \mathbb{I}^+ \cup \{0\} \\ \frac{1+5\log_e x}{3} & x \in 2n\pi + \frac{3\pi}{2} & n \in \mathbb{I}^+ \cup \{0\} \\ 1 & x \in (2n\pi + \frac{3\pi}{2}, 2n\pi + 2\pi) & n \in \mathbb{I}^+ \cup \{0\} \end{cases}$$

$f(x)$ is discontinuous at all positive odd integral

multiple of $\frac{\pi}{2}$

$f(x)$ is continuous at integral multiple of π

12. Ans (A,B,D)

$$\text{Total functions} = 3^4 = 81$$

Now into functions = Total function-onto function

$$= 81 - \frac{4!}{1!1!2!} \times \frac{1}{2!} \times 3!$$

$$= 81 - 36 = 45$$

$$(A) \quad x + y + z = 11 : (x, y, z \in \mathbb{N})$$

$$x' + y' + z' = 8$$

$$\therefore {}^{10}C_2 = 45 \quad (A) \text{ is correct}$$

$$(B) \quad {}^{10}C_8^2 \Rightarrow {}^{10}C_2 = 45 \Rightarrow (B) \text{ is correct}$$

$$(C) \quad 3!4! = 144 \Rightarrow (C) \text{ is incorrect}$$

$$(D) \quad 3600 = 2^4 \cdot 3^2 \cdot 5^2$$

$$\text{Total divisions} = 5 \times 3 \times 3 = 45$$

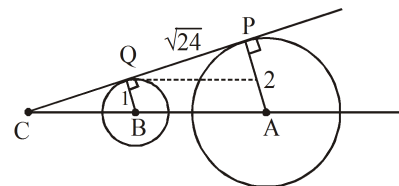
$$(D) \text{ correct}$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (7.34 or 7.35)

Point B is (6, 0), AB = 5



$$\text{area of quadrilateral ABQP} = \frac{1}{2}(2+1)\sqrt{24} = 3\sqrt{6}$$

2. Ans (0.25)

$$n(E_2) = {}^9C_2, n(E_3) = {}^9C_3$$

$$P\left(\frac{E_1}{E_2}\right) = \frac{P(E_1 \cap E_2)}{n(E_2)} = \frac{18}{{}^9C_2} = \frac{1}{2}$$

$$P\left(\frac{E_1}{E_3}\right) = \frac{P(E_1 \cap E_3)}{P(E_3)} = \frac{n(E_1 \cap E_3)}{n(E_3)} = \frac{42}{{}^9C_3} = \frac{1}{2}$$

$$\left. \begin{aligned} &\begin{bmatrix} 0 & 0 & 0 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \Rightarrow 6 \\ &\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \Rightarrow 2({}^3C_1 \times 2 \times 2) = 24 \\ &\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 1 \\ 1 & 1 & 1 \end{bmatrix} \Rightarrow 2({}^3C_1 \times 2) = 12 \end{aligned} \right\} 42$$

3. Ans (2.65 or 2.66)

$$|1 + b\omega + c\omega^2|^2 = \frac{1}{2} [(1-b)^2 + (b-c)^2 + (c-1)^2]$$

$$|1 + b\omega + c\omega^2|^2 \geq 3, \text{ when } b = 1, c = 2$$

$$A(1), B(\omega), C(2\omega^2)$$

$$\text{area of } \triangle ABC \text{ is } = \frac{5\sqrt{3}}{4}$$

4. Ans (9.00)

$$\text{For } P(1, 2) \quad t_1 = 1$$

$$\text{and for } Q(4, 4) \quad t_2 = 2$$

∴ Point of intersection of tangents is

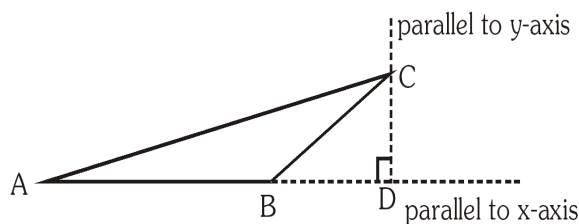
$$A(at_1t_2, a(t_1 + t_2))$$

∴ it is A(2, 3)

$$\text{Mid point of P and Q is } B\left(\frac{5}{2}, 3\right)$$

∴ line joining midpoint of P and Q to point of intersection of tangents is parallel to x-axis.

$$\therefore \text{base of the triangle ABC is } AB = \frac{5}{2} - 2 = \frac{1}{2}$$



∴ Point of intersection of normals is

$$C(2a + a(t_1^2 + t_2^2 + t_1t_2), -at_1t_2(t_1 + t_2))$$

∴ it is C(9, -6)

$$\Rightarrow CD = 9$$

$$\Rightarrow \text{Area of triangle ABC} = \frac{1}{2} \cdot AB \cdot CD$$

$$= \frac{1}{2} \cdot \frac{1}{2} \cdot 9 = \frac{9}{4}$$

$$\Rightarrow 4A = 9$$

5. Ans (55.78)

Suppose

$$\left| \frac{\frac{n(n+1)}{2} \cdot n(n-1) \cdot 2^{n-2} + n \cdot 2^{n-1}}{n \cdot 2^{n-1} \cdot 4^n} \right| + \frac{n(n+1)}{2} \cdot 4^n = 0$$

$$\frac{n(n+1)}{2} \cdot 4^n - n^2(n-1) \cdot 2^{2n-3} - n^2 2^{2n-2} + n \frac{(n+1)}{2} 4^n = 0$$

$$n(n+1) - \frac{n^2(n-1)}{8} - \frac{n^2}{4} = 0$$

$$n = 8$$

$$\begin{aligned} \text{Now } \sum_{k=1}^8 \frac{{}^8C_k}{k+1} &= \sum_{k=1}^8 \frac{k+1}{9} \cdot {}^9C_{k+1} \cdot \frac{1}{k+1} \\ &= \frac{1}{9} \cdot [{}^9C_2 + {}^9C_3 + {}^9C_4 + {}^9C_5 + {}^9C_6 + \dots + {}^9C_9] \\ &= \frac{1}{9} = \frac{1}{9} [2^9 - 10] = \frac{502}{9} = 55.78 \end{aligned}$$

6. Ans (5.00)

$$\begin{aligned} I &= \int_0^{\frac{\pi}{2^{n+1}}} \frac{\sqrt{\cos(2^n \theta)}}{(\sqrt{\cos 2^n \theta} + \sqrt{\sin(2^n \theta)})^t} d\theta \\ I &= \int_0^{\frac{\pi}{2^{n+1}}} \frac{\sqrt{\cos(2^n (\frac{\pi}{2^{n+1}} - \theta))}}{(\sqrt{\cos 2^n (\frac{\pi}{2^{n+1}} - \theta)} + \sqrt{\sin 2^n (\frac{\pi}{2^{n+1}} - \theta)})^5} d\theta \\ 2I &= \int_0^{\frac{\pi}{2^{n+1}}} \frac{d\theta}{(\sqrt{\cos 2^n \theta} + \sqrt{\sin 2^n \theta})^4} \\ 2I &= \int_0^{\frac{\pi}{2^{n+1}}} \frac{\sec^2 2^n \theta}{(1 + \sqrt{\tan 2^n \theta})^4} d\theta \end{aligned}$$

$$\text{Let } \tan 2^n \theta = t^2$$

$$2^n \sec^2 2^n \theta d\theta = 2t dt$$

$$2^n I = \int_0^\infty \frac{t dt}{(1+t)^4}$$

$$= \int_0^\infty \left(\frac{1}{(1+t)^3} - \frac{1}{(1+t)^4} \right) dt$$

$$= \left[\frac{1}{-2(1+t)^2} + \frac{1}{3(1+t)^3} \right]_0^\infty$$

$$\left[0 - \left(-\frac{1}{2} + \frac{1}{3} \right) \right] = \frac{1}{6} \Rightarrow I = \frac{1}{3(2)^{n+1}}$$



CLASSROOM CONTACT PROGRAMME

(Academic Session : 2021 - 2022)

JEE (Advanced)

AIOT

14-08-2022

JEE(Main + Advanced) : ENTHUSIAST & LEADER COURSE : (SCORE ADVANCED)

ANSWER KEY

PAPER-2

PART-1 : PHYSICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	A,D	B,C	A,D	B,C,D	B,D	B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	87.20 to 87.40	60.00	10.00	22.50	371.00	8.30
SECTION-III	Q.	1	2	3	4	5	6
	A.	3	5	1	6	9	3

PART-2 : CHEMISTRY

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,C	A,B,D	A,B	A,B,D	B,C	A,B,C,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	2.00	2.00	3.00	6.00	0.02	0.40
SECTION-III	Q.	1	2	3	4	5	6
	A.	5	7	5	6	2	1

PART-3 : MATHEMATICS

SECTION-I	Q.	1	2	3	4	5	6
	A.	B,D	A,C	A,C	A,B,C,D	A,B,D	B,D
SECTION-II	Q.	1	2	3	4	5	6
	A.	8.40	-0.25	1.50	1.00	0.70	1.50
SECTION-III	Q.	1	2	3	4	5	6
	A.	5	4	5	0	7	5

HINT – SHEET

PART-1 : PHYSICS

SECTION-I

1. Ans (A,D)

$$\lambda = \frac{h}{mv}$$

$$\frac{dh}{dt} = -\frac{h}{mv^2} \frac{dV}{dt}$$

$$10^{-4} = -\frac{10^{-34} \times 6.6}{6.6 \times 10^{-30}} \frac{1}{V^2} \frac{dV}{dt}$$

$$\frac{dV}{dt} = -V^2$$

$$\int_{10}^v \frac{dV}{dt} = - \int_0^t dt$$

$$\frac{1}{V} - \frac{1}{10} = t$$

$$\frac{1}{V} = \frac{1+10t}{10}$$

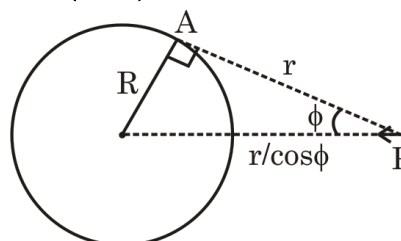
$$V = \frac{10}{1+10t}$$

$$\text{at } t = 0.9 \text{ s } V = 1 \text{ m/s}$$

$$\text{Also } a = -V^2$$

$$|a| = 1 \text{ m/s}^2$$

2. Ans (B,C)



$$V_A = V_C = \frac{kp}{(r/\cos \phi)^2} = V_P + V_{\text{induced}}$$

3. Ans (A,D)

$$C_1 = 4\pi\epsilon_0 r_1 = 4\pi\epsilon_0 R$$

$$C_2 = \frac{4\pi\epsilon_0 r_2^2}{d} = \frac{4\pi\epsilon_0 4R^2}{R/50} = 800\pi\epsilon_0 R$$

If charges appearing on the conducting spheres

'A' and 'B' are q_1 and q_2 respectively.

$$\frac{q_1}{q_2} = \frac{C_1}{C_2} = \frac{4\pi\epsilon_0 R}{800\pi\epsilon_0 R}$$

$$\Rightarrow \frac{q_1}{q_2} = \frac{1}{200}$$

$$\Rightarrow q_2 = 200q_1 \quad \dots(1)$$

Now, $q_1 + q_2 = Q$

$$q_1 + 200q_1 = Q$$

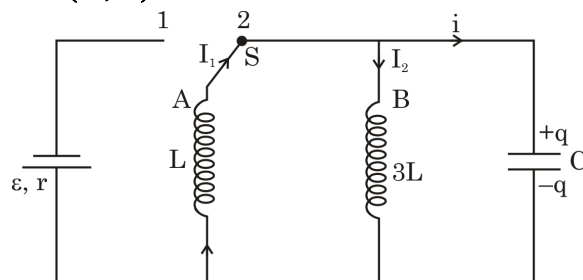
$$\Rightarrow q_1 = \frac{Q}{201}$$

$$q_2 = \frac{200Q}{201}$$

4. Ans (B,C,D)

The probability of decay in 1 sec is the decay constant λ , which remains constant. After average life 37% of nuclei remain undecayed. Part of energy of disintegration is taken by the recoiling daughter nucleus.

5. Ans (B,D)



$$I_0 = \frac{\epsilon}{r}$$

$$L \frac{dI_1}{dt} + 3L \frac{dI_2}{dt} = 0$$

$$\int_0^{t_1} dI_1 = -3 \int_0^{t_2} dI_2$$

$$I_1 - I_0 = -3I_2$$

$$I_1 + 3I_2 = I_0 \quad \dots(i)$$

$$I_1 - I_2 = I \quad \dots(ii)$$

When the charge on the capacitor is maximum,

$$I = 0$$

$$I_1 = I_2 \quad \dots(iii)$$

From equation (i) and (iii)

$$4I_0 = I_0 \Rightarrow I_2 = \frac{I_0}{4} \text{ and } I_1 = \frac{I_0}{4}$$

$$\text{Now, } \frac{1}{2} L t_0^2 - \frac{1}{2} (L + 3L) \left(\frac{t_0}{4} \right)^2 = \frac{q_{\max}^2}{2C}$$

$$L I_0^2 - \frac{L t_0^2}{4} - \frac{q_{\max}^2}{C}$$

$$\Rightarrow \frac{3L}{4} I_0^2 - \frac{q_{\max}^2}{C}$$

$$\Rightarrow q_{\max} = \frac{I_0}{2} \sqrt{3LC} = \frac{\epsilon}{2r} \sqrt{3LC}$$

6. Ans (B,D)

$$\text{Path difference} = 2D \left(1 + \frac{d^2}{D^2} \right)^{1/2} = \frac{d^2}{D}$$

$$\text{and } \frac{d_{\min}^2}{D} = \frac{\lambda}{2} \Rightarrow d_{\min} = \sqrt{\frac{\lambda D}{2}}$$

For bright fringe at x,

$$AO'P - ABP = 0$$

$$\Rightarrow D + \sqrt{D^2 + x^2} - \left(\sqrt{D^2 + d^2} + \sqrt{D^2 + (x-d)^2} \right) = 0$$

$$\Rightarrow D + D \left(1 + \frac{x^2}{2D^2} \right) - D \left(1 + \frac{d^2}{2D^2} \right) - D \left(1 + \frac{(x-d)^2}{2D^2} \right) = 0$$

$$\Rightarrow \frac{x^2}{2D} - \frac{d^2}{2D} - \frac{x^2 + d^2 - 2xd}{2D} = 0$$

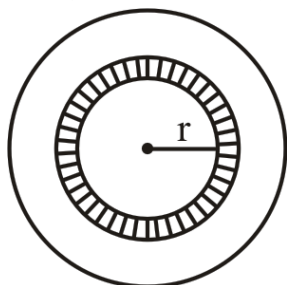
$$\Rightarrow 2xd = 2d^2$$

$$\Rightarrow x = d_{\min}$$

PART-1 : PHYSICS

SECTION-II

1. Ans (87.20 to 87.40)



Acceleration due to gravity at radial distance r

$$g(r) = \frac{G \left(\frac{4}{3} \pi r^3 \rho \right)}{r^2}$$

$$g(r) = \frac{4\pi G \rho r}{3}$$

pressure at radial distance r

$$dP = -(dr)\rho g = -\frac{4\pi G \rho r^2}{3} dr$$

$$P = \frac{-4\pi \rho^2 G}{3} \int_R^r r dr = \frac{-4\pi \rho^2 G}{3} \left[\frac{r^2}{2} \right]_R^r$$

$$P = \frac{2\pi \rho^2 G}{3} (R^2 - r^2)$$

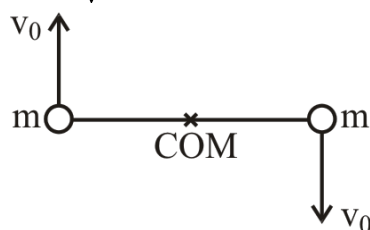
at centre of earth r = 0

$$P = \frac{2\pi}{3} G R^2 \rho^2 = 87.3 \text{ kPa}$$

2. Ans (60.00)

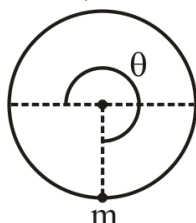
Just after collision,

$$v_0 = \sqrt{2gh} = 6 \text{ m/s}$$



Velocity of COM is zero at this moment. After this COM is in free fall. w of balls about COM

$$\omega = \frac{v_0}{\ell/2} = \frac{2v_0}{\ell}$$



Angle turned in time t,

$$\theta = \omega t = \frac{2v_0 t}{\ell} = \frac{3\pi}{2} \text{ rad}$$

COM will fall down by a distance y during time t,

$$y = \frac{1}{2} g t^2$$

Hence distance between ball and point of collision

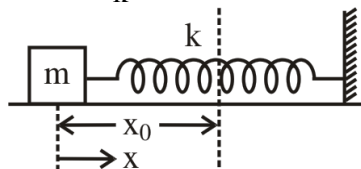
$$\Rightarrow y + \frac{\ell}{2} = \frac{1}{2} \times 10 \times 0.06 + 0.3$$

$$\Rightarrow y = 60 \text{ cm}$$

3. Ans (10.00)

Elongation in the spring when the block begins to slide,

$$x_0 = \frac{\mu_s mg}{k}$$



If the displacement of block is x from the point where it started sliding then

$$k(x_0 - x) - \mu_k mg = ma$$

$$\Rightarrow ma = -kx + kx_0 - \mu_k mg$$

$$\Rightarrow a = -\frac{k}{m}x + (\mu_s - \mu_k)g$$

$$\text{If } x = x' + \frac{m}{k}(\mu_s - \mu_k)g$$

$$\Rightarrow a = \frac{-k}{m}x'$$

Motion will be simple harmonic (valid only till the time block stops).

$$\omega = \sqrt{\frac{k}{m}} \text{ amplitude } A = \frac{m}{k}(\mu_s - \mu_k)g$$

$$\begin{aligned} \therefore v_{\max} &= \omega A = \sqrt{\frac{m}{k}}(\mu_s - \mu_k)g \\ &= \sqrt{\frac{2}{200}} \times 1 = 10 \text{ cm/s} \end{aligned}$$

4. Ans (22.50)

The resistive force is proportional to velocity

$$ma = -kv$$

$$\Rightarrow \frac{mdv}{dt} + kv = 0$$

$$\Rightarrow \frac{d}{dt}(mv + kx) = 0$$

$$\Rightarrow \frac{d}{dt}\left(\frac{m}{k}v + x\right) = 0$$

$$\Rightarrow \frac{m}{k}v + x = C \quad (\text{Constant}) \quad \dots(1)$$

If initial velocity is v_0 and maximum stopping

distance is $x_{\max}$

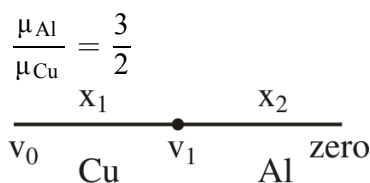
$$C = \frac{mv_0}{k} = x_{\max} \quad \dots(2)$$

$$\Rightarrow x_{\max} = \mu v_0 \quad \dots(3)$$

where $\mu = \frac{m}{k}$ is the characteristic of the metal

plate causing the breaking.

Hence $\mu_{Al}v_0 = 30$ cm and $\mu_{Cu}v_0 = 20$ cm



Now, $\mu v + x = C = \mu v_0$

$$\Rightarrow \mu v_1 + 15 = \mu v_0$$

$$v_1 = v_0 - \frac{15}{\mu_{Cu}}$$

$$\text{and } x_2 = \mu_{Al}v_1 = \mu_{Al}\left(v_0 - \frac{15}{\mu_{Cu}}\right)$$

$$\begin{aligned} \Rightarrow x_2 &= \mu_{Al}v_0 - \frac{\mu_{Al}}{\mu_{Cu}} \times 15 \\ &= 30 - \frac{3}{2} \times 15 = 7.5 \text{ cm} \end{aligned}$$

$$x_1 + x_2 = 15 + 7.5 = 22.5 \text{ cm}$$

5. Ans (371.00)

$$V_B = 2V_C, P_B = 2P_C$$

$$V_P = \frac{3}{2}V_C, P_P = \frac{3}{2}P_C, T_P = \frac{9}{4}T_C = 630 \text{ K},$$

$$C_P = \frac{5}{2}R, C_V = \frac{3}{2}R$$

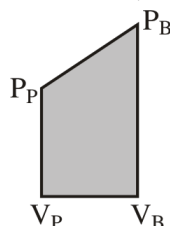
$$Q_{AP} = nC_P \Delta T = 2 \times \frac{5R}{2} (630 - 280)$$

$$Q_{AP} = 1750 R$$

$$\text{From } P \rightarrow B, \Delta U = nC_V \Delta T = 2 \times \frac{3R}{2} \times (1120 - 630)$$

$$\Delta U = 1470 R$$

Work done,



$$W = \frac{1}{2} \times (P_P + P_B) \times (V_B - V_P)$$

$$W = \frac{7}{32} nRT_B = 490 R$$

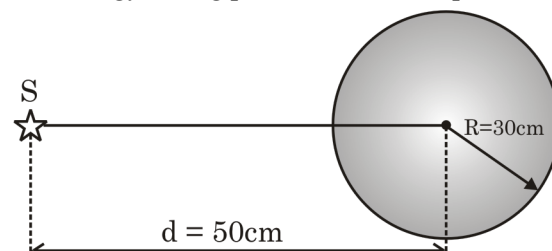
$$\therefore Q_{PB} = \Delta U + W = (1470 + 490)R$$

$$Q_{PB} = 1960 R$$

$$\text{Total heat} = 1960 R + 1750 R = 3710 R$$

6. Ans (8.30)

The energy falling per second on the sphere is



$$\begin{aligned} P &= \frac{P_0}{4\pi} \times 2\pi(1 - \cos \theta) = \frac{P_0}{2} \left(1 - \frac{4}{5}\right) \\ &= \frac{P_0}{10} = \frac{0.4}{10} = 4 \times 10^{-2} W \end{aligned}$$

Energy of each photon is

$$E = 5 \times 1.6 \times 10^{-19} = 8 \times 10^{-19} \text{ J}$$

$\therefore$ Number of photons falling per second on the sphere.

$$N = \frac{P}{E} = \frac{4 \times 10^{-2}}{8 \times 10^{-19}} = 5 \times 10^{16} \text{ photons/sec}$$

$\therefore$ Number of photoelectrons emitted per second

$$n = \frac{5 \times 10^{16}}{10^6} = 5 \times 10^{10} \text{ photoelectrons/sec}$$

PART-1 : PHYSICS

SECTION-III

1. Ans (3)

Momentum of inertia of the triangular wire frame about OO' is

$$I = \lambda \ell \left(\frac{\sqrt{3}}{2} \ell \right)^2 + \frac{\lambda \ell}{3} \left(\frac{\sqrt{3}}{2} \ell \right)^2 + \frac{\lambda \ell}{3} \left(\frac{\sqrt{3}}{2} \ell \right)^2$$

$$I = \frac{5}{4} \lambda \ell^3$$

Magnetic dipole moment of wire triangle

$$|\vec{m}| = \text{current} \times \text{Area} = \frac{\sqrt{3}}{4} i \ell^2$$

$$\text{Torque } |\vec{\tau}| = |\vec{m} \times \vec{B}| = \frac{\sqrt{3}}{4} Bi \ell^2$$

$$\text{Angular acceleration } \alpha = \frac{|\vec{\tau}|}{I} = \frac{\frac{\sqrt{3}}{4} Bi \ell^2}{\frac{5}{4} \lambda \ell^3} = \frac{\sqrt{3} Bi}{5 \lambda \ell}$$

Linear acceleration of the horizontal side,

$$a = \alpha R = \frac{\sqrt{3} Bi}{5 \lambda \ell} \cdot \frac{\sqrt{3}}{2} \ell = \frac{3}{10} \frac{Bi \ell}{\lambda} = \frac{n Bi \ell}{10 \lambda}$$

$$\therefore n = 3$$

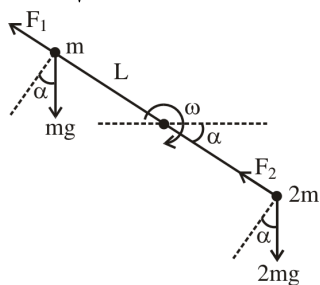
2. Ans (5)

Moment of inertia of the system = 9 mL^2

By energy conservation

$$\frac{1}{2} I \omega^2 = (2mg)(2L \sin \alpha) - mgL \sin \alpha$$

$$\omega = \sqrt{\frac{2g \sin \alpha}{3L}}$$



F_1 = force on m due to rod

F_2 = force on $2m$ due to rod

$$mg \sin \alpha - F_1 = mL \omega^2$$

$$\Rightarrow F_1 = \frac{1}{3} mg \sin \alpha$$

$$\text{and } F_2 - 2mg \sin \alpha = (2m)(2L)\omega^2$$

$$\Rightarrow F_2 = \frac{14}{3} mg \sin \alpha$$

Total force by rod on the axis = $F_1 + F_2$

$$= \frac{1}{3} mg \sin \alpha + \frac{14}{3} mg \sin \alpha = 5mg \sin \alpha$$

$$n = 5$$

3. Ans (1)

$$\text{Initial height of water} = \frac{\text{vol}}{\text{area}} = \frac{5 \text{ dm}^3}{10 \text{ dm}^2}$$

$$h_1 = 0.5 \text{ dm} = 5 \times 10^{-2} \text{ m}$$

Pressure at the base due to liquid

$$P_1 = h_1 \rho_1 g = 5 \times 10^{-2} \times 10^3 \times 10 \text{ Pa}$$

$$P_1 = 500 \text{ Pa}$$

$$P = \frac{\text{Weight}}{\text{Area}}$$

$$\frac{\Delta P}{P} = \frac{-\Delta A}{A} = \frac{-2\gamma_{\text{Steel}}}{3} \Delta T = -2 \times \frac{5}{3} \times 10^{-4} \times 60$$

$$\Delta P = -2 \times 10^{-2} P = -2 \times 10^{-2} \times 500 \text{ Pa}$$

$$\Delta P = -1 \text{ Pa}$$

4. Ans (6)

$$\int_0^{V_0} C V^2 dV = 1 = \text{Total probability}$$

$$\Rightarrow \frac{C V_0^3}{3} = 1$$

$$\Rightarrow C = \frac{3}{V_0^3} = \frac{x}{V_0^y}$$

$$\therefore x + y = 6$$

5. Ans (9)

$$\log_2 \left(\frac{P}{P_0} \right) = 1$$

$$\left(\frac{P}{P_0} \right) = 2$$

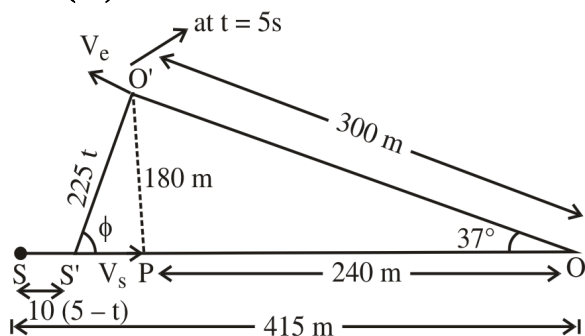
According to Stefan's Law $P \propto T^4$

$$\left(\frac{P_2}{P_1} \right) = \left(\frac{T_2^4}{T_1^4} \right)$$

$$\left(\frac{P_2}{P_0} \right) = 2 \times 4^4 = 2^9$$

$$\therefore \log_2 \left(\frac{P_2}{P_0} \right) = 9$$

6. Ans (3)



Let the sound released at the position S' reaches the observer at $t = 5s$.

Let t be the time taken by sound to reach observer from S'.

$$S'O = 225t$$

$$S'S = 10(5 - t)$$

$$S'P = (415) - (240) - 10(5 - t)$$

$$= 415 - 240 + 50 + 10t = 125 + 10t$$

$$\text{Also, } (125 + 10t)^2 + (180)^2 = (225t)^2 \Rightarrow t = 1s, \phi = 53^\circ$$

$$f' = \frac{f_v}{V - V_s \cos \phi} = 675 \text{ Hz}$$

PART-2 : CHEMISTRY

SECTION-I

5. Ans (B,C)

$$\Delta S_{\text{source}} = \frac{dq_{\text{rev}}}{T} = \frac{-1000}{500} = -2 \text{ J/K}$$

$$\Delta S_{\text{sink}} = \frac{800}{300} = \frac{8}{3} \text{ J/K}$$

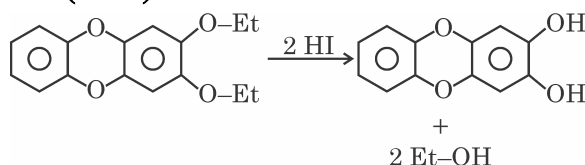
$$\Delta S_{\text{Total}} = \frac{2}{3} \text{ (Irreversible engine)}$$

$$\eta = 1 - \frac{T_C}{T_H} = 1 - \frac{3}{5} = 0.4$$

PART-2 : CHEMISTRY

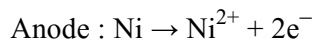
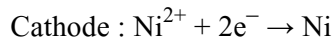
SECTION-II

1. Ans (2.00)



5. Ans (0.02)

When Ni electrodes are used, reactions involved are :



So there is no net change in concentration.

6. Ans (0.40)

Using 1st equation :

$$P_{\text{CO}_2} = K_P = 8 \times 10^{-2}$$

Using 2nd equation

$$K_P = \frac{P_{\text{CO}}^2}{P_{\text{CO}_2}} \Rightarrow 2 = \frac{P_{\text{CO}}^2}{0.08}$$

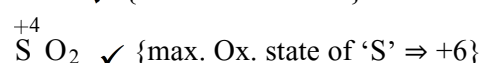
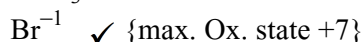
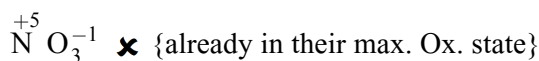
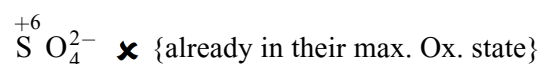
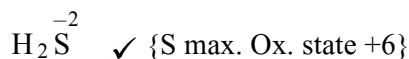
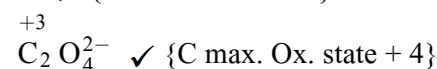
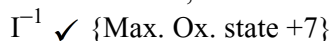
$$\Rightarrow P_{\text{CO}} = 0.4 \text{ atm.}$$

PART-2 : CHEMISTRY

SECTION-III

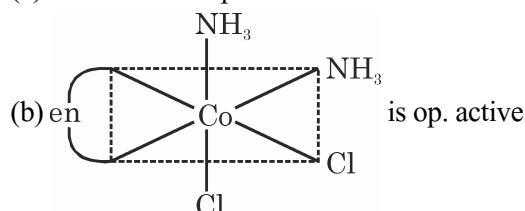
3. Ans (5)

The species which are not in 'their maximum oxidation state, can be oxidised by KMnO_4 .



4. Ans (6)

(a) Cis-isomer is op. active



(c) optically inactive.

(d) op. inactive

(e) always optically active

(f) square planar, optically inactive

(g) EDTA complexes are optically active

(h) 3 didentate ligands are always optically active

(i) same as (h)

5. **Ans (2)**

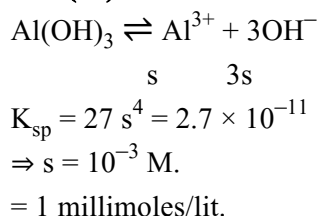
$$(k_1 + k_2) = \frac{x}{t}$$

$$3 \times 10^{-2} = \frac{x}{100}$$

$$\Rightarrow x = 3 \text{ M}$$

$$[C] = 3 \times \frac{2}{3} = 2 \text{ M}$$

6. **Ans (1)**



PART-3 : MATHEMATICS

SECTION-I

1. **Ans (B,D)**

$$f(x) = \ln^2 x - \frac{1}{6} \Rightarrow f(e) = \frac{5}{6}$$

$$f'(x) = \frac{2 \ln x}{x} \geq 0 \quad \forall x \in [1, \infty)$$

$f(x)$ is strictly increasing function

$$f(1) = -\frac{1}{6} < 0$$

$$f''(x) = 2 \left[\frac{1 - \ln x}{x^2} \right]$$

Equation of tangent :

$$y - \frac{5}{6} = \frac{2}{e}(x - e) \Rightarrow y = \frac{2x}{e} - \frac{7}{6}$$

$$\text{Area} = \int_1^e \left[\left(\ln^2 x - \frac{1}{6} \right) - \left(\frac{2x}{e} - \frac{7}{6} \right) \right] dx = e + \frac{1}{e} - 3$$

2. **Ans (A,C)**

For non trivial solution

$$\Rightarrow \begin{vmatrix} 1 & -c & -b \\ c & -1 & a \\ b & a & -1 \end{vmatrix} = 0 \Rightarrow 1 - 2abc - a^2 - b^2 - c^2 = 0$$

$$\Rightarrow a^2 + b^2 c^2 + 2abc = 1 - b^2 - c^2 + b^2 c^2$$

$$\Rightarrow (a + bc)^2 = (1 - b^2)(1 - c^2)$$

$$\text{Similarly ; } (b + ac)^2 = (1 - a^2)(1 - b^2)$$

$$(c + ab)^2 = (1 - a^2)(1 - b^2)$$

So $1 - a^2$, $1 - b^2$, $1 - c^2$ are of same sign and atleast one in proper fraction, so all are positive.

$$\Rightarrow 1 - a^2 > 0, 1 - b^2 > 0; 1 - c^2 > 0 \Rightarrow a^2 + b^2 + c^2 < 3$$

$$\text{and } 1 - 2abc < 3 \Rightarrow abc > -1$$

3. **Ans (A,C)**

$$\left. \begin{array}{l} a > b, a > c > d \\ a > b > c > d \\ a > c > b > d \\ a > c > d > b \end{array} \right\} {}^9C_4 \times 3 = 378$$

$$\left. \begin{array}{l} a > b = c > d \\ a > c > b = d \end{array} \right\} {}^9C_3 \times 2 = 168$$

$$\therefore \text{Total numbers} = 378 + 168 = 546$$

4. **Ans (A,B,C,D)**

$$h(x) = \begin{cases} f(x) & ; 0 \leq x^2 < \pi^2 \\ \frac{f(x)+g(x)}{2} & ; x^2 = \pi^2 \\ g(x) & ; x^2 > \pi^2 \end{cases}$$

$$h(x) = \begin{cases} g(x) & ; x < -\pi \\ \frac{f(-\pi)+g(-\pi)}{2} & ; x = -\pi \\ f(x) & ; x \in (-\pi, \pi) \\ \frac{f(\pi)+g(\pi)}{2} & ; x = \pi \\ g(x) & ; x > \pi \end{cases}$$

If $h(x)$ is continuous at $x = \pi$, then

$$1 - \alpha + 2 = \beta = \frac{1 - \alpha + 2 + \beta}{2}$$

$$\therefore \alpha + \beta = 3$$

5. **Ans (A,B,D)**

$$\frac{x^2}{3} + y^2 = 1, \text{ if } x = \sqrt{3} \cos \theta, y = \sin \theta$$

$$\text{Now, } E = (x + y + 3)^2 - 9$$

$$= (\sqrt{3} \cos \theta + \sin \theta + 3)^2 - 9$$

$$\text{Let } \sqrt{3} \cos \theta + \sin \theta = t$$

$$\Rightarrow f(t) = t^2 + 6t, t \in [-2, 2]$$

$$f(2) = 16, f(-2) = -8$$

$$M = 16, m = -8$$

6. Ans (B,D)

$$\lim_{x \rightarrow 0} \frac{x^2 \frac{\sin \beta x}{\beta x} \times \beta x}{\alpha n + \frac{(\alpha x)^3}{3} + \dots - \left(x - \frac{x^3}{3} + \dots\right)} = 4$$

$$\Rightarrow \lim_{x \rightarrow 0} \frac{x^3 \times \beta}{x(\alpha - 1) + \left(\frac{\alpha^3}{3} + \frac{1}{3!}\right)x^3 + \dots} = 4$$

$$\Rightarrow \alpha = 1 \text{ and } \frac{\beta}{\frac{1}{3} + \frac{1}{3!}} = 4$$

$$\Rightarrow \beta = 4 \times \left(\frac{1}{3} + \frac{1}{6}\right)$$

$$= 4 \times \frac{1}{2} = 2$$

PART-3 : MATHEMATICS

SECTION-II

1. Ans (8.40)

Case-I

$$\begin{array}{|c|c|} \hline \text{---} & \frac{1}{2} \\ \hline \text{two odd} & \frac{1}{6} \\ \hline \text{and 1 even} & \frac{2}{4} \\ \hline \end{array}$$

Case-II

$$\begin{array}{|c|c|} \hline \text{---} & \frac{2}{4} \\ \hline \text{all are} & \frac{2}{6} \\ \hline \text{odd} & \frac{4}{4} \\ \hline \end{array}$$

$$\begin{array}{c} 3 \quad 2 \\ 3 \quad 6 \\ 5 \quad 2 \\ 5 \quad 6 \\ {}^2C_2 \cdot {}^2C_1 \cdot 3! \times 6 + {}^3C_3 \cdot 3! \times 2 \\ = 72 + 12 \\ = 84 \end{array}$$

2. Ans (-0.25)

$$\begin{vmatrix} \alpha & \beta & \gamma \\ 1 & 1 & 0 \\ 1 & 0 & 1 \end{vmatrix} = 0 \Rightarrow \alpha - \beta - \gamma = 0$$

$$\Rightarrow \alpha = \beta + \gamma$$

$$\vec{P} = \lambda (\hat{Q} + \hat{R})$$

$$= \frac{\lambda}{\sqrt{2}} [2\hat{i} + \hat{j} + \hat{k}]$$

$$\therefore \alpha = \frac{2\lambda}{\sqrt{2}}, \beta = \frac{\lambda}{\sqrt{2}}, \gamma = \frac{\lambda}{\sqrt{2}}$$

$$\frac{3\beta - 4\gamma}{2\alpha} = -0.25$$

3. Ans (1.50)

$$f(x) = x^2 - x^2 \int_0^1 dt - \int_0^1 f^2(t) dt - 2x \int_0^1 f(t) dt$$

$$f(x) = - \int_0^1 f^2(t) dt - 2x \int_0^1 f(t) dt$$

$$f'(x) = -2 \int_0^1 f(t) dt = \text{constant} \Rightarrow f(x) \text{ is linear}$$

$$\therefore f(x) = ax + b$$

$$a = -2 \int_0^1 (at + b) dt \quad \dots(1)$$

$$b = - \int_0^1 (at + b)^2 dt \quad \dots(2)$$

$$\text{from (1) \& (2) } a = 3 ; b = -3$$

$$\therefore f(x) = 3x - 3$$

$$\text{Area bounded by } f(x) \text{ with } x = 0$$

$$\text{and } y = 0 \text{ is } \frac{3}{2} = 1.50 \text{ square units.}$$

4. Ans (1.00)

$$|z_1^2 - z_2^2| = |\bar{z}_1^2 + \bar{z}_2^2 - 2\bar{z}_1\bar{z}_2|$$

$$|z_1^2 - z_2^2| = |(z_1 - z_2)^2|$$

$$\Rightarrow |z_1 + z_2| = |z_1 - z_2|$$

$$z_1 \perp z_2$$

$$\Rightarrow \arg\left(\frac{z_1}{z_2}\right) = 2n\pi \pm \frac{\pi}{2} \Rightarrow \frac{-\pi}{2}, \frac{\pi}{2}$$

$$\text{Minimum value of } \left|\frac{a-2b}{3}\right| \text{ when } a = 1 \text{ and}$$

$$b = 2$$

5. Ans (0.70)

$$\int_2^3 \left(\frac{\frac{3}{x^5} - \frac{1}{x}}{1 - \frac{1}{x^4}} \right) dx = \int_2^3 \left(\frac{\frac{3}{x^7} - \frac{1}{x^3}}{\frac{1}{x^2} - \frac{1}{x^6}} \right) dx$$

$$\text{put } \frac{1}{x^2} - \frac{1}{x^6} = t \Rightarrow \left(-\frac{2}{x^3} + \frac{6}{x^7} \right) dx = dt$$

$$\frac{1}{2} \int_{\frac{15}{64}}^{\frac{80}{729}} \frac{dt}{t} = \frac{1}{2} \ln \left(\frac{2^{10}}{3^7} \right)$$

$$\text{Hence, } n = 7 ; m = 10$$

6. Ans (1.50)

$$I_{2022+i} - I_i = \int_0^1 \frac{x^i (x^{2022} - 1)}{x^{2022} - 1} dx = \frac{1}{i+1}$$

$$J_{2026+j} + J_{j+3} = \int_0^1 \left(\frac{x^{2026+j}}{x^{2023} + 1} + \frac{x^{j+3}}{x^{2023} + 1} \right) dx$$

$$= \int_0^1 x^{j+3} \frac{(x^{2023} + 1)}{(x^{2023} + 1)} dx$$

$$= \frac{1}{j+4}$$

$$\therefore A = \begin{bmatrix} \frac{1}{2} & 0 & 0 \\ 0 & \frac{1}{3} & 0 \\ 0 & 0 & \frac{1}{4} \end{bmatrix}; B = \begin{bmatrix} \frac{1}{5} & 0 & 0 \\ 0 & \frac{1}{6} & 0 \\ 0 & 0 & \frac{1}{7} \end{bmatrix}$$

$$\frac{1}{20} \det(B^{-1}) - \text{trace}(A^{-1}) = \frac{30 \times 7}{20} - 9$$

$$= \frac{3}{2} = 1.50$$

PART-3 : MATHEMATICS

SECTION-III

1. Ans (5)

$$H: \frac{x^2}{9} - \frac{y^2}{4} = 1$$

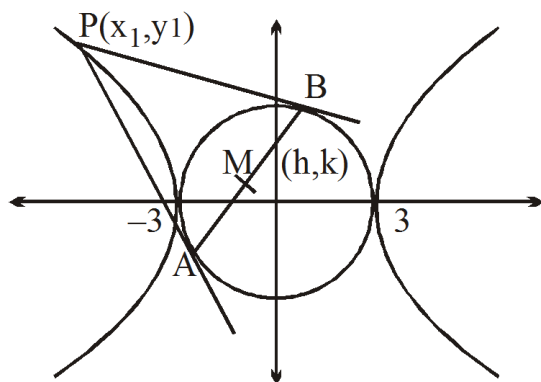
$$\text{Equation of AB: } xx_1 + yy_1 = 9 \quad \dots\dots(1)$$

(Chord of contact)

$$xh + yk = h^2 + k^2 \quad \dots\dots(2) \quad (T = S_1)$$

On comparing equation (1) and (2), we get

$$\frac{x_1}{h} = \frac{y_1}{k} = \frac{9}{h^2 + k^2}$$



$$x_1 = \frac{9h}{h^2 + k^2} = 3 \sec \theta$$

$$y_1 = \frac{9k}{h^2 + k^2} = 2 \tan \theta$$

On using $\sec^2 \theta - \tan^2 \theta = 1$, we get

$$\left(\frac{3h}{h^2 + k^2} \right)^2 - \left(\frac{9k}{2(h^2 + k^2)} \right)^2 = 1$$

$$\Rightarrow \frac{81x^2}{9} - \frac{81y^2}{4} = (x^2 + y^2)^2$$

$$\Rightarrow \frac{x^2}{9} - \frac{y^2}{4} = \left(\frac{x^2 + y^2}{9} \right)^2$$

$$\therefore a = 9; b = 4$$

2. Ans (4)

	(1)	(2)	(3)
Red balls	20%	40%	40%
Black balls	25%	45%	30%

$$P = \frac{\frac{15}{35} \times \frac{40}{100}}{\frac{15}{35} \times \frac{40}{100} + \frac{20}{35} \times \frac{45}{100}}$$

$$P = \frac{40}{40 + 60} = \frac{2}{5}$$

3. Ans (5)

$$|z - 2022| = |z + 2022|$$

$\Rightarrow z$ lies on imaginary axis i.e. $z + \bar{z} = 0$

Let $A(3, -4)$ and $B(5, 20)$ and 'P' be any point on imaginary axis

$$|z - 3 + 4i| + |z - 5 - 20i| \text{ is minimum}$$

$\Rightarrow PA + PB$ is minimum.

Let $A'(-3, -4)$ (reflection of A in imaginary axis) join $A'B$.

Its equation is $y = 3x + 5$.

$$\text{Hence } P(0, 5) \Rightarrow z = 5i$$

$$z^2 = -25$$

$$\therefore \lim_{\theta \rightarrow 0} \frac{\sin(10\theta)}{\tan^{-1}(2\theta)} = 5$$

4. Ans (0)

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{5^n} {}^n C_r \left(\sum_{t=0}^{r-1} {}^r C_t 2^t \right)$$

$$\lim_{n \rightarrow \infty} \sum_{r=1}^n \frac{1}{5^n} {}^n C_r [3^r - 2^r]$$

$$\lim_{n \rightarrow \infty} \frac{[4^n - 3^n]}{5^n} = 0$$

5. Ans (7)

$$A = (I + B)(I - B)^{-1}$$

$$AA^T = (I + B)(I - B)^{-1}((I - B)^T)^{-1}(I + B)^T$$

$$= (I + B)(I - B)^{-1}(I + B)^{-1}(I - B)$$

$$= (I + B)((I + B)(I - B))^{-1}(I - B)$$

$$= (I + B)((I - B)(I + B))^{-1}(I - B)$$

($\because I + B, I - B$ are commutative)

$$\therefore AA^T = (I + B)(I + B)^{-1}(I - B)^{-1}(I - B) = I$$

$$\therefore AA^T = I$$

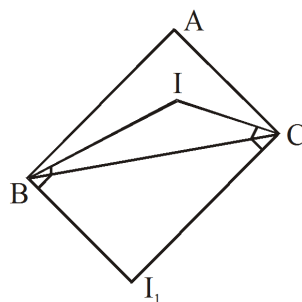
$$|AA^T| = |I| = 1 \Rightarrow |A|^2 = 1$$

$$\text{as } |A| > 0 \Rightarrow |A| = 1$$

$$\det(2A) - \det(\text{adj}(A)) = 8 \det(A) - \det(A)^2$$

$$= 8(1) - (1)^2 = 7.$$

6. Ans (5)



Circumcentre of BIC is mid point of I and I_1 .

where I is $(-2, 3)$ and I_1 is $(-6, -2)$

$$(2b - a) = 5$$